

CSCE 640 QUANTUM ALGORITHMS · MODULE 02 · LECTURE NOTES

States and Measurement

What a qubit holds, and what you are allowed to see.

MODULE OUTCOMES

- Write and manipulate single- and multi-qubit states.
- Explain the measurement rule and its consequences.
- Explain entanglement without mysticism.
- State the no-cloning result and why it matters.
- Write a state vector simulator.

1 The state

$$|\text{psi}\rangle = a|0\rangle + b|1\rangle, \quad \text{with } |a|^2 + |b|^2 = 1$$

- A single qubit is two complex amplitudes, normalised — and it is not a probability pair: a and b may be negative or complex, and that is the entire point (see the callout).
- For n qubits, $|\text{psi}\rangle = \sum_x a_x |x\rangle$ over all x in $\{0,1\}^n$ — 2^n amplitudes indexed by the classical bit strings, and the state space is the tensor product of n two-dimensional spaces rather than n copies of one.
- A global phase is unobservable: $e^{i\theta}|\text{psi}\rangle$ is the same physical state as $|\text{psi}\rangle$, always — while the relative phase between terms is observable and is where everything happens (Module 04 §1).
- This course writes kets as $|0\rangle$ in ASCII throughout, for typographic reasons: the conventional angle-bracket glyphs do not render reliably in the toolchain that produced these notes, and a visible notation is worth more than a conventional one.

A SUPERPOSITION IS NOT A PROBABILITY DISTRIBUTION, AND THE DIFFERENCE IS THE NEGATIVE NUMBERS

A probabilistic bit is 0 with probability p and 1 with probability $1 - p$ — and probabilities cannot cancel, because they are non-negative and sum to one.

A qubit's amplitudes can be negative or complex — so two computational paths arriving at the same outcome can sum to zero, which is a resource a randomised algorithm simply does not have (CSCE 658's whole machinery is built without it).

Which is why the right comparison is not 'quantum is random' but 'quantum is random with cancellation' — and the cancellation is the entire advantage (Module 04 states this as the course's central mechanism).

And it is also why a classical simulation is expensive: you cannot sample your way through it, because the amplitudes that matter are precisely the ones that will later cancel — so a Monte Carlo approach that follows likely paths misses the interference.

2 Measurement

- Measuring in the computational basis returns the string x with probability $|a_x|^2$, and leaves the state as $|x\rangle$ — everything else is gone irrecoverably.
- So you get n bits per run out of 2^n amplitudes — which is the fundamental bottleneck of the whole subject, and the reason every algorithm must concentrate amplitude onto the answer before measuring.
- And the phases are invisible to measurement — $|a_x|^2$ discards them entirely — so phase has to be converted into amplitude by interference first, which is what every algorithm in Modules 05 through 08 is doing.
- Measurement is irreversible, unlike every other operation in the model — which is why it comes last, and why measuring an intermediate qubit is a design decision with consequences rather than a free diagnostic.
- And repeating the whole computation does not help directly: each run samples the same distribution, so you learn the distribution, not the state — and learning a distribution over 2^n outcomes takes exponentially many samples. n bits out of 2^n amplitudes: the whole art of quantum algorithm design is making those n bits worth having.

3 Entanglement

product state: $(|0\rangle + |1\rangle)(|0\rangle + |1\rangle)/2 = (|00\rangle + |01\rangle + |10\rangle + |11\rangle)/2$

entangled: $(|00\rangle + |11\rangle)/\sqrt{2}$

- A product state factors into one state per qubit — measuring one qubit tells you nothing about the other, and the description of the whole is just the descriptions of the parts.
- An entangled state does not factor — measuring the first qubit of $(|00\rangle + |11\rangle)/\sqrt{2}$ determines the second — and that correlation is the whole content of the term.
- So the test is: does the 2^n -dimensional vector factor as a tensor product? If not, it is entangled. It is a linear-algebra property of the vector, not a physical influence travelling between qubits — and nothing in the formalism moves.
- Defining entanglement as non-factorisability removes essentially all the mystery, and it also makes the computational role clear: **entanglement is how a circuit builds amplitudes that depend jointly on many qubits**, which is what makes the interference in Module 04 possible at all.

AND NO-CLONING, WHICH IS A ONE-LINE CONSEQUENCE WITH LARGE EFFECTS

There is no unitary U with $U(|\psi\rangle|0\rangle) = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$ — and the proof is two lines from linearity: suppose it works for $|0\rangle$ and for $|1\rangle$, then apply it to their superposition and compare. **It is one of the very few results in the subject you can derive on sight.**

Which closes the obvious workaround to §2's bottleneck: **you cannot copy the state a thousand times and measure each copy in a different basis** in order to read the amplitudes out.

And it is what makes quantum key distribution possible — **an eavesdropper cannot copy the channel** — which is the one quantum technology that is actually deployed rather than projected, and is worth distinguishing from quantum computing when people conflate them.

Plus it constrains error correction severely — **the classical trick of storing three copies and taking a majority is simply unavailable** — which is why Module 10's codes are cleverer than repetition and why fault tolerance took a decade to establish.

4 Write the simulator

```

import numpy as np

# state: complex vector of length 2**n, index = bit string
def zero_state(n):
    v = np.zeros(2**n, dtype=complex); v[0] = 1.0
    return v

# apply a 2x2 gate U to qubit q of an n-qubit state
def apply1(v, U, q, n):
    v = v.reshape([2]*n)
    v = np.moveaxis(v, q, 0)
    v = np.tensordot(U, v, axes=([1], [0]))
    return np.moveaxis(v, 0, q).reshape(2**n)

# measure everything, once
def measure(v, rng):
    p = np.abs(v)**2
    return rng.choice(len(v), p=p / p.sum())

```

Controlled gates are the same idea with an index mask — apply the gate only to the amplitudes whose control bit is 1 — **and that is the entire simulator. The amplitudes stop being abstract once you have written this**, which is why **Project 1 asks for exactly this, written rather than imported**. Verify it against a hand calculation on two qubits including an entangled state, because a simulator that handles product states correctly and entangled ones incorrectly looks right until it does not.

AND THE EXPONENTIAL COST IS THE POINT, NOT A LIMITATION OF YOUR CODE

The vector has 2^n entries, so 30 qubits is about 16 gigabytes in double-precision complex — which is why a laptop stops at around 25 to 30 qubits of fully general state, and no amount of optimisation changes the exponent.

And that cost is exactly the reason the subject exists — Feynman's observation was that simulating quantum systems classically appears to be exponentially hard, so use a quantum system to do it (Module 09, and Module 01 §3's first item).

So your simulator's limit is informative rather than annoying: it is the same wall that makes quantum simulation worth doing on a quantum device, encountered personally on a laptop, which is a better way to understand it than reading the claim.

Though note the honest qualification — states with special structure simulate efficiently: low entanglement (tensor-network methods), Clifford-only circuits (the Gottesman-Knill theorem), and shallow circuits — which Module 13 §3 shows is precisely how several published advantage claims were reduced after the fact.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Nielsen & Chuang, chapters 2 and 4 (library copy)	§§1 to 3 — the linear algebra and the measurement postulates, developed carefully.
Wootters & Zurek — A single quantum cannot be cloned	§3's callout in the original — one page, and the argument is the one you can reconstruct.
Watrous's notes, the early lectures (free)	§§1 and 2 , rigorous and short — and the measurement formalism is stated more generally than Nielsen-Chuang does at first.
Aaronson — Multilinear formulas and the Gottesman-Knill theorem, lecture notes (free)	§4's qualification — which circuit classes simulate classically, and why that matters for advantage claims.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write three single-qubit states** and verify normalisation.
- 2 **Show that a global phase changes no measurement probability.**
- 3 **Construct two amplitudes that cancel**, and say what the probabilistic analogue would be.
- 4 **Compute the measurement distribution** for a three-qubit state by hand.
- 5 **Show that measurement discards relative phase.**
- 6 **Factor three two-qubit states**, or show that they do not factor.
- 7 **Prove no-cloning** from linearity, in your own words.
- 8 **Write the simulator** from Part 4, and verify it on an entangled state by hand.
- 9 **Find your simulator's qubit ceiling** on your own machine, and report the memory.
- 10 **Simulate a Clifford-only circuit** and read about why it could have been done classically in polynomial time.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Write a general n -qubit state and count its parameters.
- 2 Which phase is observable, and which is not?
- 3 Why is a superposition not a probability distribution?
- 4 State the measurement rule and three consequences.

- 5 How many bits per run, and why does that matter?
- 6 Why must phase be converted to amplitude?
- 7 Define entanglement, and say what it is not.
- 8 Prove no-cloning, and give two of its effects.
- 9 What is the memory cost of n qubits, and where is the ceiling?
- 10 Name three classically simulable circuit classes.